

Physics-informed neural network model for transient wave propagation in a pressurized pipeline

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Abstract

This paper explores the use of neural networks (NNs) to model water-hammer waves propagation in a bounded pipe system. The training dataset is obtained from a numerical solution using the Method Of Characteristics (MOC). It is found that conventional NNs fail to model the water-hammer waves propagation when trained using only the boundary conditions and require large datasets for training. However, the use of physics-informed NN (PINN), wherein the one-dimensional wave equation, along with the boundary conditions are imposed as part of the loss function, provides accurate results. The PINN model parameters (i.e., weights and biases) are optimized using (i) the Adaptive Moment Estimation (ADAM), and (ii) the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithms. The results show that L-BFGS based solution is in good agreement with the exact MOC solution, whereas the ADAM-based solution has slow convergence. Moreover, it is found that the PINN solution is sensitive to changes in boundary and initial conditions as well as to the time length of the wavefield solution. In particular, our results show that as the time length (or bandwidth) of the signals increases, a denser NN model architecture (in terms of number of neurons) is required to accurately predict the response. It is also found that the PINN model is sensitive to the location of additional measurement points used as a constraining dataset. Detailed sensitivity analysis concerning model architecture and the constraining dataset is yet to be investigated. The advantages and implications of using PINN in hydraulic transient modeling are discussed.

Keywords: Artificial intelligence; Leak detection; Hydraulics; Machine learning; Water-hammer

1. INTRODUCTION

Recently, Raissi et al., (2019) proposed an elegant framework, named physics-informed neural networks (PINNs), to predict the solution of differential equations using neural networks (NNs). As the name implies, this approach unifies data-driven machine learning and domain knowledge to optimize the machine learning (ML) model. PINNs have gained significant attention in the scientific community and particularly in computational society due to their remarkable ability to predict solutions for PDEs. Another substantial advantage of PINN is that this approach requires significantly less data for training in comparison to the conventional NN models due to the fact that physical laws are enforced as part of the loss function in the optimization procedure (Karniadakis et al., 2021). Raissi et al., (2019) demonstrated the strength of PINNs by finding the solutions for the Burger's equation, 1D Schrödinger equation, and Allen-Cahn equation. Waheed et al., (2021) used PINNs to solve the eikonal equation to predict the seismic wave travel times for seismological applications. Lai et al., (2021) used the PINN framework to solve the ODEs for structural identification applications using numerical as well as experimental data. Huang et al., (2021) extended the PINN framework to account for the point sources in the PDEs.

Motivated by these recent intriguing developments of PINN, this work uses PINN for water hammer application and discusses its potential and possible use in practice. Unlike other systems of interest (e.g., open-channel flows, or structures), pipeline systems pose different and unique challenges. That is, these systems are underground, hence poorly accessible, have partially known topologies, and are comprised of dynamic boundary conditions (BCs). Under these constraints, the adoption of the PINN framework to predict the transient response of buried pipe systems is not a straightforward task. For instance, it is impossible to measure the initial conditions (ICs) continuously along the pipe in a functioning or operating pipe system. This paper shows that additional measurement points can compensate for the missing IC constraint. In what follows, the methodology is presented at first. Then, the numerical examples are provided. In particular, the solution of the PINN is compared with conventional NN and discussed. The sensitivity of the PINN model concerning the measurement

locations, initial conditions (ICs), time-extent of the response, and frequency of the wave source are explored. Finally, the conclusions and future perspectives are discussed.

2. METHODOLOGY

Consider a reservoir-pipe-reservoir system. The pipe is assumed to be straight, horizontal, and frictionless for simplicity. A pulse-type forcing is imposed from the downstream end by maneuvering the downstream boundary. The transient response of this system is governed by the following one-dimensional water-hammer wave equation (Ghidaoui, 2004):

$$\frac{\partial^2 h}{\partial t^2} - a^2 \frac{\partial^2 h}{\partial x^2} = 0 \quad [1]$$

where $h(x, t)$ is the pressure head, a is the wave speed, $t \in [0, T]$ is the time, $x \in [0, L]$ denotes the space, with L and T being the total length of the pipe, and the total time of the experiment, respectively. The initial condition (ICs) and boundary conditions (BCs) are

$$\text{BCs:} \quad h(t, x=0) = H_R; \quad t > 0 \quad [2]$$

$$h(t, x=L) = S(t); \quad t > 0 \quad [3]$$

$$\text{ICs:} \quad h(t=0, x) = H_0(x); \quad 0 \leq x \leq L \quad [4]$$

$$q(t=0, x) = Q_0(x); \quad 0 \leq x \leq L \quad [5]$$

where

$$S(t) = H_0(L) + \exp\left[-\frac{1}{2}\left(\frac{t - \mu_T}{\sigma_T}\right)^2\right]; \quad t > 0 \quad [6]$$

wherein H_R is the steady-state pressure head at the upstream end, $S(t)$ is the Gaussian-type excitation $\mu_T = 0.5(L/a)$, $\sigma_T = 0.1(L/a)$ (unless specified otherwise), $H_0(x)$ & $Q_0(x)$ are the steady-state pressure head & flowrate, respectively. In this paper, Eq. [1] is solved using MOC to obtain the constraining dataset for PINN models.

2.1 Physics-informed Neural Networks (PINNs)

Figure 1 shows the PINN architecture used in this paper. The network is comprised of 5 layers each of which contains 20 neurons (unless specified otherwise). The inputs of the PINN model are N number of collocation points for space $x_f \in [0, L]$ and time $t_f \in [0, T]$. The output of the model is $h^{\text{PINN}}(x, t)$. The loss function is defined as

$$Loss_T = Loss_D + Loss_P \quad [7]$$

wherein

$$Loss_P = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| f(t_f^i, x_f^i) \right|^2 \quad [8]$$

$$f(x_f, t_f) := \frac{\partial^2 h}{\partial t^2} - a^2 \frac{\partial^2 h}{\partial x^2} \quad [9]$$

$$Loss_D = Loss_{IC} + Loss_{BC} \quad [10]$$

$$Loss_{IC} = \left\| h^{\text{Data}} \Big|_{t=0} - h^{\text{PINN}} \Big|_{t=0} \right\|_{\text{MSE}} + \left\| q^{\text{Data}} \Big|_{t=0} - q^{\text{PINN}} \Big|_{t=0} \right\|_{\text{MSE}} \quad [11]$$

$$Loss_{BC} = \left\| h^{\text{Data}}(x=0) - h^{\text{PINN}}(x=0) \right\|_{\text{MSE}} + \left\| h^{\text{Data}}(x=L) - h^{\text{PINN}}(x=L) \right\|_{\text{MSE}} \quad [12]$$

where $\|\cdot\|_{\text{MSE}}$ represents the mean-squared error, $Loss_T$ is the total loss, $Loss_D$ is the data-based loss which includes: (i) boundary conditions ($Loss_{BC}$) & (iii) initial conditions ($Loss_{IC}$), $Loss_P$ denotes the physics-based loss

function, $f(x_f, t_f)$ is the residue of the wave equation, N_f is a subset of N and represents the total number of $\{x_f, t_f\}$ inputs involved in each iteration. Since the output of the PINN model is the pressure head, $q^{\text{PINN}}(t=0)$ is not known. This problem is alleviated by providing an additional dataset from one measurement location from inside the pipe domain (if one IC is unknown) or two measurement points (if both ICs are unknown). More analysis and discussion on this subject are provided in Section 3.2. In this case, $Loss_D$ can be redefined as

$$Loss_D = Loss_{IC} + Loss_{BC} + Loss_{XM} \quad [13]$$

where

$$Loss_{XM} = \sum_{m=1}^M \|h^{\text{Data}}(t, x_m) - h^{\text{PINN}}(t, x_m)\|_{\text{MSE}} \quad [14]$$

$Loss_{IC}$ includes the available IC, $x_m \in (0, L)$ is the spatial coordinate of additional measurement point, and M is the total number of additional measurement points. For the case where both ICs are unknown, $Loss_{IC} = 0$ in Eq. [13]. The spatial and temporal derivatives involved in Eq. [9] are computed using the automatic differentiation (AD) technique (Baydin et al., 2018). To optimize the model parameters (i.e., weights of neurons w , biases b , and activation functions σ), two optimization algorithms are used: (i) Adaptive Moment Estimation (ADAM) with mini-batches (Kingma and Ba, 2014), and (ii) Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithm with full-batch.

3. NUMERICAL EXAMPLES

Consider a reservoir-pipe-reservoir system of length $L = 100$ m, diameter $D = 0.1$ m, and wave speed $a = 1000$ m.s⁻¹. The steady-state pressure head and flow rate in the system are $H_0 = 20$ m and $Q_0 = 4$ l.s⁻¹, respectively. The numerical simulations are performed using the MOC. The transient wave is generated by perturbing the pressure head at the downstream end in the form of a Gaussian pulse (Eq. [3]). The pipe section is discretized with $\Delta x = 2$ m and the temporal discretization is $\Delta t = 0.002$ s. The total simulation time is $T = 4L/a = 0.4$ s which includes one complete wave period. The simulated wavefield solution is shown in Figure 2 wherein both x and y axes are respectively non-dimensionalized with respect to L and L/a , and the color pattern shows the non-dimensionalized pressure head of the system with respect to Joukowski's pressure (Wylie and Streeter, 1993).

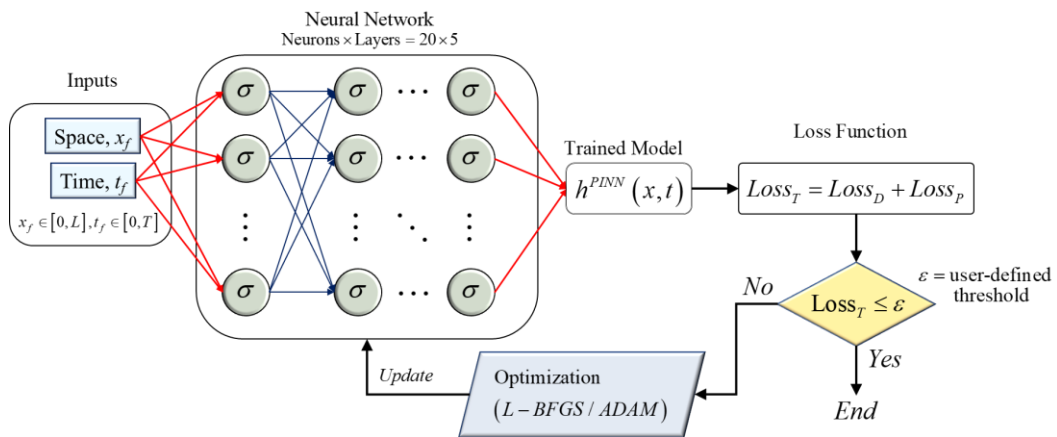


Figure 1. Physics-informed Neural Network architecture with five layers (1 input, 3 hidden, and 1 output) each of which contains 20 neurons.

The input data for space x and time t is obtained by generating 20×10^3 collocation points from the Sobol set with the threshold of $[0, T]$ for time and $[0, L]$ for space. For L-BFGS, the total number of iterations and maximum function evaluation are set to 30×10^3 . The optimality threshold is set to 1×10^{-5} . For the ADAM optimizer, the number of epochs is set to 7500 with the minibatch size of 5000 such that the total number of iterations required to complete the epochs is 30×10^3 . For both algorithms, the hyperbolic tangent function “tanh” is used as an activation function. The choice of these hyperparameters (i.e., number of layers, size of each

layer, number of iterations and epochs, and activation function) is made with a trial-and-error approach. The model is optimized on a CPU with the following specifications: Intel® Xeon® W-2295 CPU@ 3.00 GHz with 256 GB RAM.

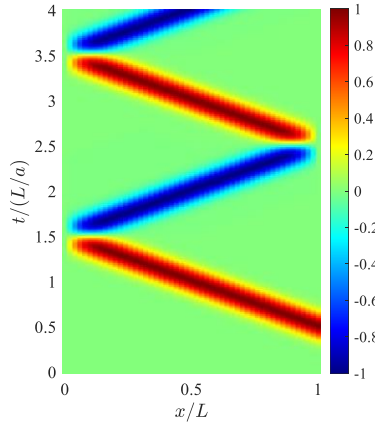


Figure 2. Exact MOC solution in space-time diagram.

3.1 Results

In the initial analysis, the PINN model was optimized with two boundary conditions, and two pressure-head profiles at $t = 0$ and $t = \Delta t$ as ICs (the latter IC is provided to compensate for Eq. [5]). It is found that PINN model is unable to predict the transient response of the system. Those initial results are not shown here for brevity. However, by providing the steady-state pressure head profile (at $t = 0$, i.e., Eq. [4]) and one dataset from $x_m / L = 0.3$ (i.e., $Loss_{XM}$) gives satisfactory results. Therefore, in what follows, the PINN models are optimized using two BCs, one IC (pressure head profile), and one additional measurement from $x_m / L = 0.3$ (unless specified otherwise). The scenario wherein both ICs are unknown is also discussed in Section 3.2.

3.1.1 Performance comparison between ADAM and L-BFGS algorithms

Figure 3 gives the loss functions of both optimizers and shows that L-BFGS has a faster convergence rate in comparison to ADAM. In terms of computational cost, the ADAM optimizer was completed in about 3 hours and 30 minutes, while for the same number of iterations (i.e., for 30×10^3 iterations), L-BFGS took around 4 hours and 54 minutes to complete. Studies suggest that ADAM optimizer with a mini-batch approach requires more epochs (i.e., more iterations) to achieve effective results which may increase the computational cost Raissi et al., (2019). However, as known for any machine learning model, these suggestions are highly problem/application dependent. An appealing approach to reduce the computational cost is to adopt the transfer learning approach suggested by Raissi et al., (2019) wherein a pre-optimized model can be used to re-optimize over a new dataset.

In terms of accuracy, the results obtained with each optimizer are presented in Figure 4. The intensity of the color in this figure shows the amplitude of the non-dimensional pressure head. Figure 4(a) displays the exact MOC solution. Figure 4(b) shows that L-BFGS has accurately predicted the transient response of the system at all time steps. In comparison, Figure 4(c) shows that the ADAM optimizer has roughly captured the overall wave path and needs more iterations to further refine the model parameters. Similar observations are also noticed by other researchers in this area (e.g., Alkhalifah et al., (2021); Tartakovsky et al., (2020); Raissi et al., (2019)). Therefore, in what follows, only L-BFGS algorithm is used.

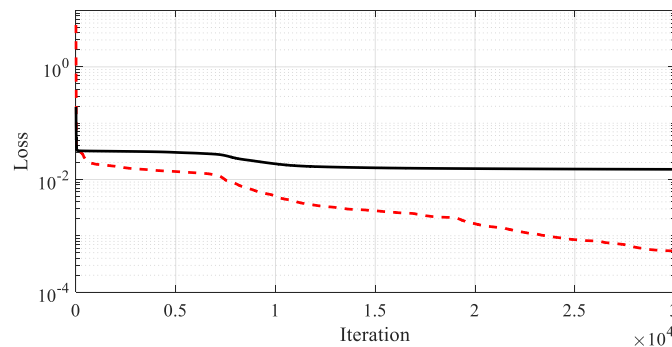


Figure 3. Loss function versus the number of total iterations for L-BFGS (dashed line) and ADAM (solid line).

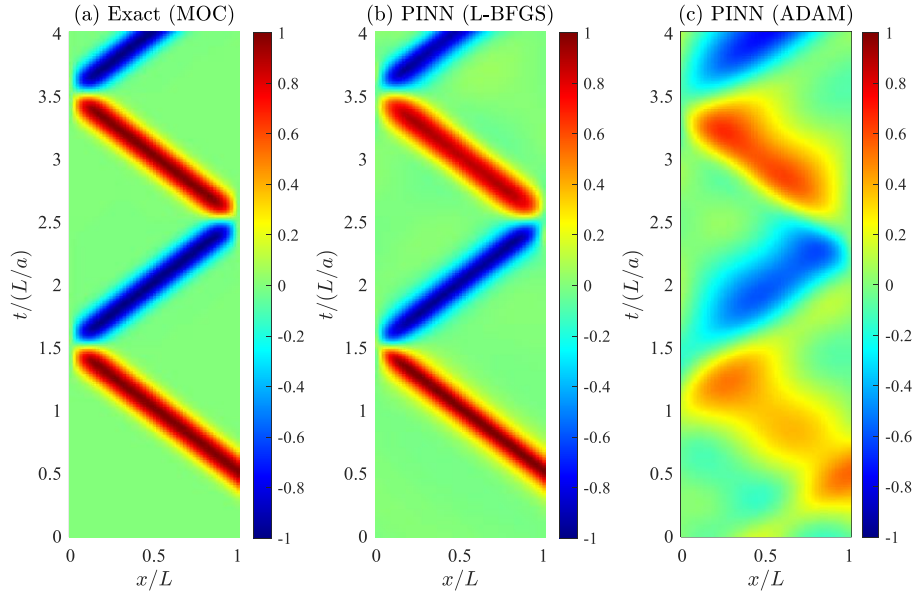


Figure 4. Space-time diagram of wave propagation in (a) exact MOC, (b) PINN with L-BFGS optimizer, and (c) PINN with ADAM optimizer.

3.1.2 Detailed analysis of PINN model optimization with L-BFGS algorithm

Figure 5 shows the timelapse of wave propagation for the first half-period $t = 2L/a$. Each box in this figure contains the exact MOC solution (solid black line) and PINN solution (dotted red line) optimized with the L-BFGS algorithm. The PINN model has successfully predicted the wave propagation for all spatial coordinates at different time steps. A closer look at these results reveals that the predicted and exact pressure profiles have minor discrepancies in terms of predicting the sharpness of the pulse. This form of discrepancies is attributed to the number of layers of the network. The original paper on PINNs by Raissi et al., (2019) showed that as the number of layers increases, the matching between the exact and PINN-based solution improves. It is worth highlighting that there are no explicit (or well-defined) steps to define the optimal PINN architecture (i.e., number of layers and number of neurons). Such discrepancies can be minimized in several ways by: (i) increasing the number of layers as increasing the number of layers helps to predict the sharper functions (Alkhalifah et al., 2021; Raissi et al., 2019), (ii) using the transfer learning approach wherein PINN model parameters (w, b, σ) of a pre-optimized PINN model for smooth wavefield can be used as an initial point for optimizing the PINN model for sharp wavefield. These options will be further explored in future papers.

Figure 6 shows the wave propagation through the space-time diagram for (a) MOC, (b) PINN, and (c) the difference between MOC and PINN. The color bars in this figure represent the non-dimensional pressure head. This figure also demonstrates the effectiveness of the PINN model in terms of capturing the boundary conditions and wave dynamics of the system. It is worth highlighting that the PINN model is semi-exact (and not mesh-dependent) in the sense that the optimized PINN model can be used for any spatial or temporal discretization. In fact, the results shown in Figure 6(b) have higher spatiotemporal resolution with $\Delta x = 0.1$ m and $\Delta t = 0.0001$ s which is much smaller than the discretization used in the MOC model. The PINN results in Figure 6(b) are downsampled to obtain Figure 6(c). Note that the model discussed in this section is referred to as the “baseline model” in the remaining part of the paper.

3.1.3 Comparison between PINN and ordinary NN models

In this section, the PINN and ordinary NN are compared to give a brief comparison in terms of learning the solution to PDE. Both models are optimized under the same network and hyperparameters settings. The results are presented in Figures 7 and 8. Figure 7 compares the exact MOC solution, NN solution, and PINN solution at all three measurement locations which are provided as the boundary constraints. It can be seen that the ordinary NN can fairly replicate the boundary conditions (7a and 7c) and partially predicted the pressure signal at $x_m/L = 0.3$. However, it fails to predict the wave signal at other locations which were not provided as boundary constraints as can be seen from Figure 8. On the other hand, PINN has well-predicted the transient response throughout the spatial as well as temporal domains. It is important to note that Figure 8c does not mean that NN cannot be used to predict wave propagation. This is because the NN simulation in Figure 8c only used

training data set at the boundaries. However, A fair comparison between conventional NN and PINN would require that the conventional NN to be trained by providing as many training datasets within the domain as the number of collocation points used for PINN. The reason why such internal training dataset was not provided to train NN is because it is impractical. In real system, such internal measurements at random spatiotemporal points (20,000 points) cannot be collected. Moreover, even by using MOC numerical simulation, it is impractical to generate millions of mesh points to obtain 20,000 training dataset points (as provided for PINN). This practical limitation does not exist in the framework of PINN since pressure measurements at collocation points are not required. PINN computes the gradients of the NN-estimated pressure at the allocation points and optimize the modeling parameters (w, b, σ) according to the imposed wave equations (see Eqs. [8] and [9]). The ability of PINN to overcome this practical limitation is precisely the main motivation of conducting this research work.

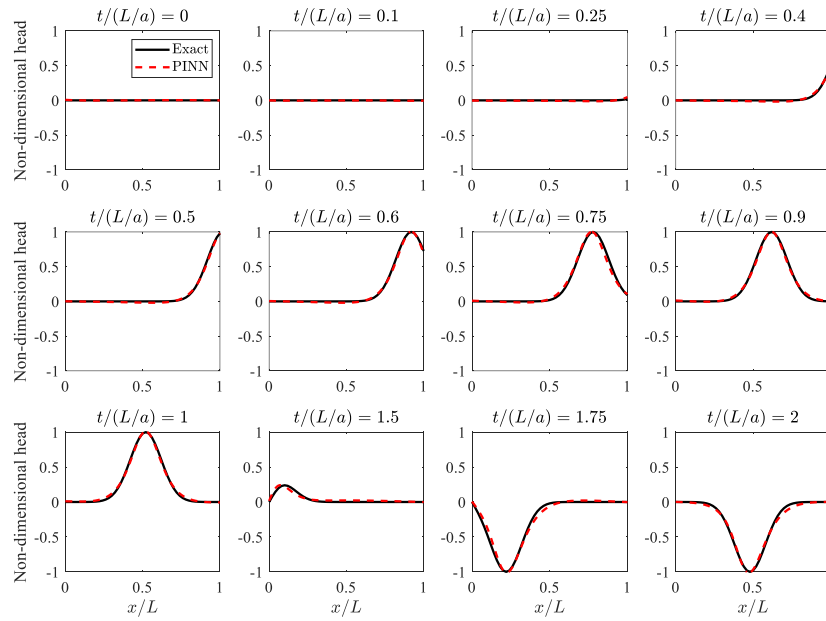


Figure 5. Comparison of MOC and PINN model in terms of timelapse of wave propagation.

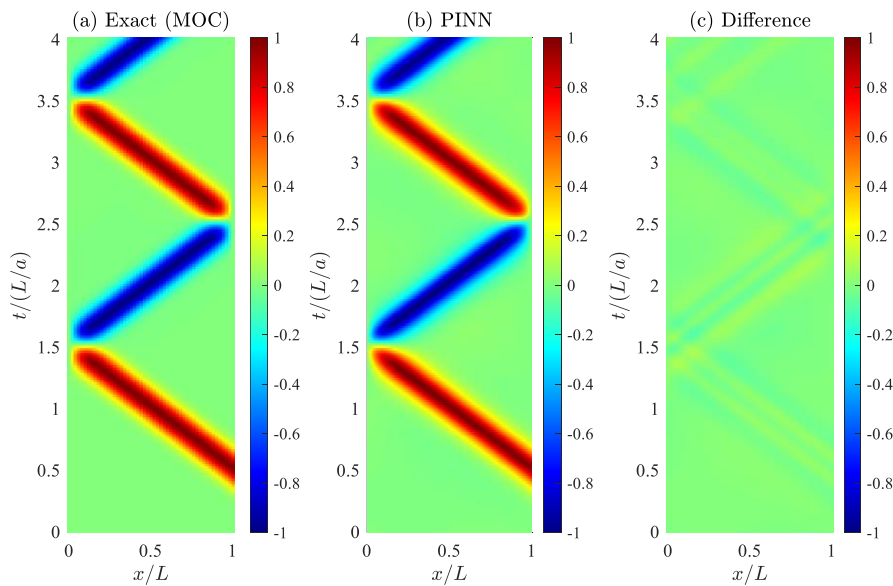


Figure 6. Space-time diagram of wave propagation in (a) Exact MOC, (b) PINN, and (c) their difference. The color bar shows the non-dimensional amplitude of the wave.

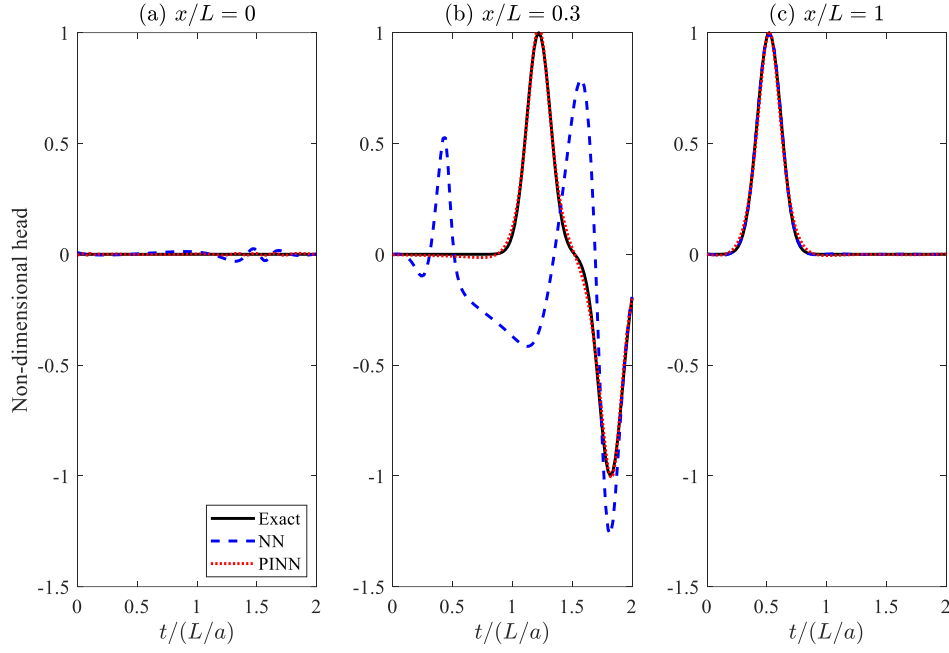


Figure 7. Pressure-time signals for MOC, NN, and PINN models at different locations.

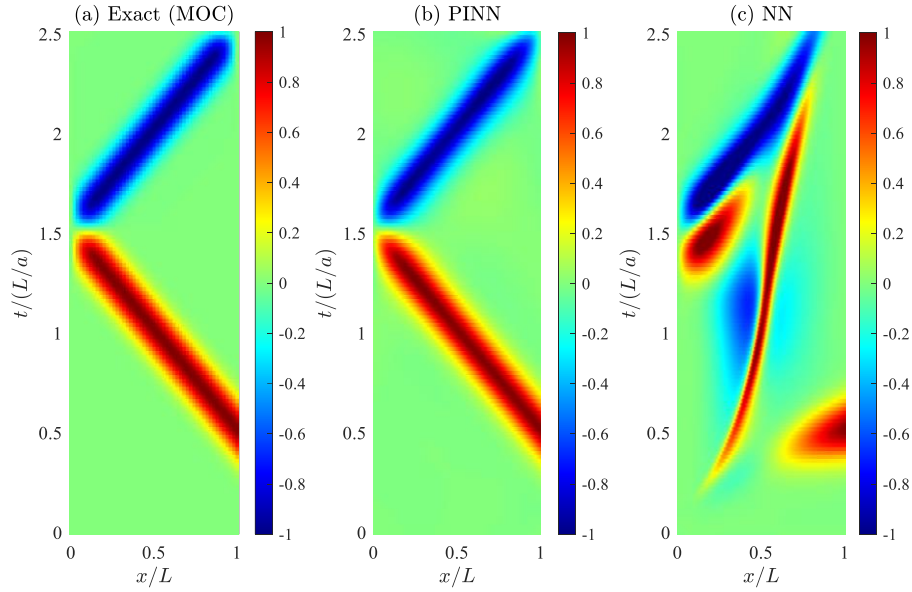


Figure 8. Space-time diagram of wave propagation in (a) MOC, (b) PINN (baseline model), and (c) ordinary NN.

3.2 Sensitivity analysis

In this section, the influence of (i) frequency of the source, (ii) time extent of the solution, and (iii) measurement locations on the accuracy and performance of PINN models are studied systematically. Figure 9 shows two wave excitations in time and frequency domains: Excitation-1 has the bandwidth of $[0, 5]$ Hz (solid line) with, $\sigma_T = 0.1(L/a)$, and is the same as the one used in the baseline model (Figure 6); Excitation-2 has the bandwidth of $[0, 10]$ Hz, with $\sigma_T = 0.04(L/a)$ (dashed line) whose exact solution is shown in Figure 10(a). The objective is to optimize the PINN model according to excitation-2 using the same PINN architecture and hyperparameters as the baseline model. Figures 10(b) and 10(c) show the performance of two PINN models: PINN-A is the solution when the model is similar to that used in the baseline model and PINN-B is the solution when the number of neurons in each layer are increased from 20 to 100. It is observed that the PINN model which successfully predicted the response subjected to excitation-1 (see Figure 6) is prone to errors when optimized according to excitation-2 that contains a sharper pressure pulse with higher frequency bandwidth (see Figure 10(b)). On the other hand, as the number of neurons in each layer are increased from 20 to 100, the

PINN model has accurately reproduced the response (see Figure 10(c)). Accurately modeling high frequency waves is an important aspect because high frequency waves are required in transient based defect detection methods (Lee et al., 2015; Waqar et al., 2022) to localize defect more accurately (higher resolution).

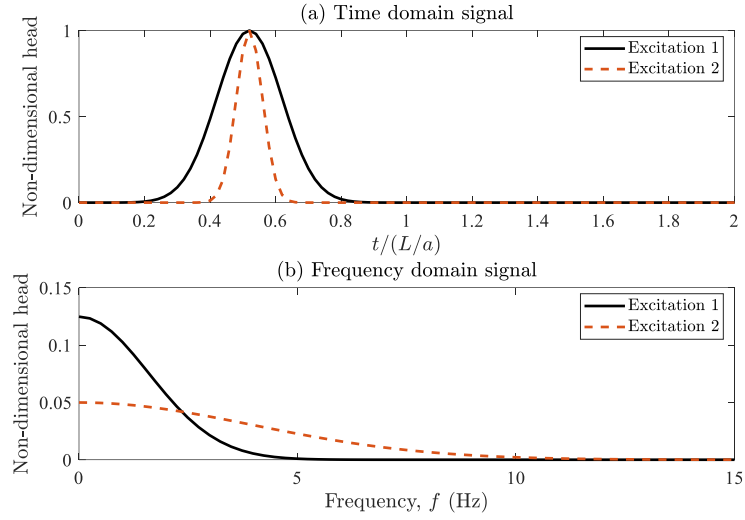


Figure 9. Space-time diagram of wave propagation in (a) MOC, (b) PINN, and (c) ordinary NN.

Regarding the time extent of the solution, Figure 11 shows the optimization of PINN models with the same initial/boundary conditions, the number of layers, and hyperparameters as described in Section 3.1 (i.e., the baseline model) but for the case where the length of the signal is $t = 8L/a$ (instead of $t = 4L/a$). In this case, two PINN models are optimized: PINN-1 with 20 neurons in each layer (i.e., same as baseline model in Section 3.1), and PINN-2 with 100 neurons in each layer. It can be seen that the PINN-1 model is able to capture only the first half-period of the wave $t = (2L/a)$, while failed to capture the remaining part of the wavefield, whereas PINN-2 has fairly predicted the transient response of the system with relatively fewer errors. The results presented in Figures 10 and 11 are intriguing and hint that the accuracy of PINNs is sensitive to the time extent of the signal as well as on the frequency content of the signal. From a practical viewpoint, the foregoing analysis implies that the adoption of the PINN framework requires a trial-and-error approach to obtain the optimal PINN architecture and hyperparameter settings.

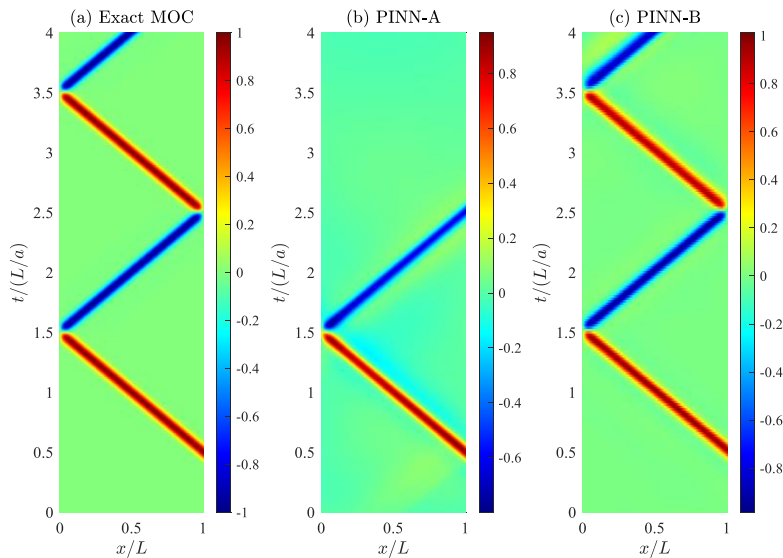


Figure 10. Space-time diagram of wave propagation in (a) Exact MOC, (b) PINN-A with Excitation 2 (20 neurons per layer), (c) PINN-B with Excitation 2 (100 neurons per layer).

Figure 12 displays the performance of 5 PINN models which have same NN model parameters as used in the baseline model but are optimized under different constraining dataset scenarios. In particular, for PINN-I, PINN-II & PINN-III, the additional measurement point (defined in Eq. [13]) is located at $x_m/L = 0.3$, $x_m/L = 0.5$

, & $x_m / L = 0.75$, respectively. The results are shown in Figures 12 (b – d) which demonstrate that PINN models also very sensitive to the measurement location of the constraining dataset. These results also show that the sensor locations which are closer to the boundaries are relatively more useful than those in the middle of the pipe. PINN-IV and PINN-V consider the scenarios when the ICs are not provided at all. In this case, PINN-IV predicts the response using only one measurement point along the pipe located at $x_m / L = 0.3$, and PINN-V predicts the response using two measurement points located at $x_m / L = [0.3, 0.5]$. The results are given in Figure 12(e-f). It is observed that providing one additional measurement location (in addition to the upstream and downstream boundary conditions) is not enough (see Figure 12(e)) as the predicted response is blurred. However, when two measurements along the pipe are provided at $x_m / L = [0.3, 0.5]$, the PINN model has accurately predicted the transient response (see Figure 12(f)). Therefore, if the initial conditions are not known (which is usually the case in practice), at least two measurement locations along the pipe are needed to optimize the PINN models.

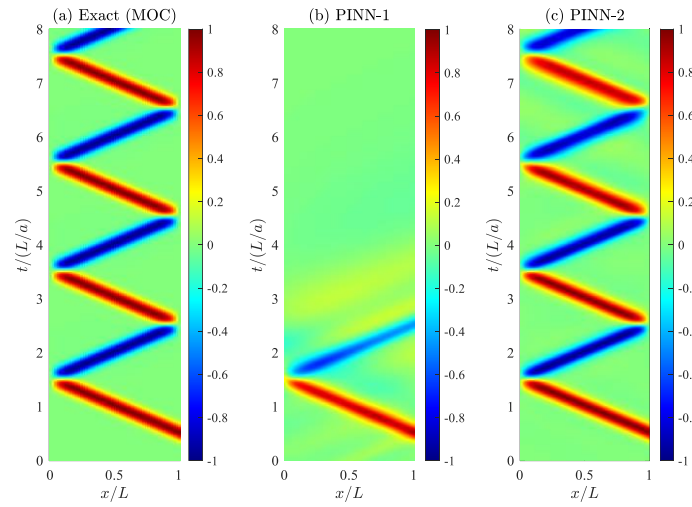


Figure 11. Space-time diagram of wave propagation in (a) MOC, (b) PINN-1 (with 20 neurons per layer), and (c) PINN-2 (100 neurons per layer).

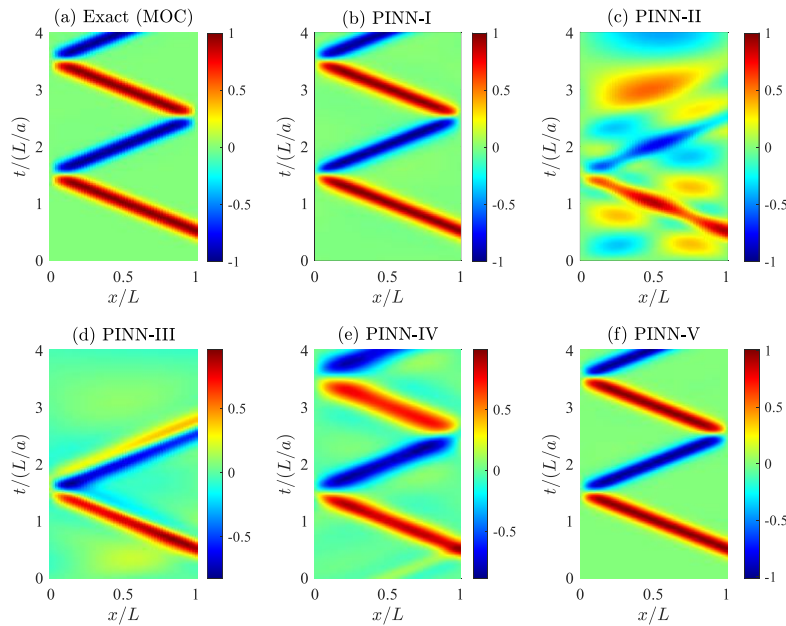


Figure 12. Space-time diagram of wave propagation in (a) Exact MOC, (b) PINN-I with $x_m / L = 0.3$ (baseline model), (c) PINN-II with $x_m / L = 0.5$, (d) PINN-III with $x_m / L = 0.75$, (e) PINN-IV with $x_m / L = 0.3$ and no IC, and (f) PINN-V with $x_m / L = [0.3, 0.5]$ and no IC.

4. CONCLUSIONS AND FUTURE PLANS

In this paper, the framework of a physics-informed neural network (PINN) is explored to predict the hydraulic transient response of a reservoir-pipe-reservoir system. The results of the PINN model are compared with the exact MOC solution and with the ordinary NN model. The results show that the PINN algorithm can predict the transient response of the system throughout the pipe domain. However, it is also noticed that the accuracy of the model is sensitive to the (i) frequency bandwidth of the signals, (ii) time length of the solution, and (iii) spatial location of data points as constraints. In particular, the results demonstrate that as the length or frequency bandwidth of the signal increases, a denser PINN architecture is required to accurately predict the response of the system. Furthermore, it is noticed that some measurement locations bring more benefits in terms of optimizing the model than others. More research is required to further elucidate the sensitivity of PINN models concerning constraining datasets and PINN architecture. Currently, the PINN framework is being extended for the case where the effect of friction and fluid-structure interaction are included (e.g., steady & unsteady frictions and viscoelastic effects). Another potential application of PINN would be to predict the transient response of a pipe network which will be explored in future work.

5. ACKNOWLEDGEMENTS

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